School of Electronic Engineering and Computer Science
Queen Mary University of London

MSc PROJECT DEFINITION
2025-26

This project definition must be undertaken in consultation with your supervisor. The feasibility of the project should have been assessed and the project aims should be clearly defined.
Submission of this document implies that you have discussed the specification with your supervisor.

Project Title: Investigating Generalisation, Limitations and Robustness in Multimodal Deepfake Detection

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PROJECT AIMS:
State the design, development or research challenge (problem) that the project aims to solve.

The aim of this project is to investigate the limitations of current deepfake detection models and methods.
This project is guided by the following research question:
To what extent do current deepfake detection models rely on dataset-specific features, and how does this affect their ability to generalise to unseen data and manipulation methods?
Many state-of-the-art detection models implicitly learn features related to specific datasets rather than generalisable features, leading to degraded performance under cross-dataset and real-world conditions.
While recent advances in generative AI models have significantly improved the quality of deepfake content, many detection methods rely on dataset-specific knowledge and fail to transfer effectively to unseen data or conditions, raising many concerns about their reliability in real-world situations.
To address this, the project will implement and evaluate recent deepfake detection methods using pretrained architectures by choosing a relevant dataset within a chosen subdomain, analysing existing state-of-the-art detection techniques, reimplementing these methods and possibly combining multiple methods to create a hybrid framework to explore whether combining these techniques can improve generalisation and robustness.
My goal is to not just evaluate performance across different conditions but also to understand the underlying reasons for why models fail in many real word scenarios and the extent to which current methods are relying on features that cannot be transferred to other data/datasets, particularly as generative AI models continue to improve, creating more room for current methods to fail in detection. The overall goal is to evaluate and improve the reliability of deepfake detection methods, as well as to provide insight into their limitations in real word scenarios.

PROJECT OBJECTIVES:
List a series of objectives you need to achieve to fulfil the aims of your project.

- Conducting a literature review of the best current deepfake detection methods and identify the key limitations of these methods.
- Selecting an appropriate dataset (e.g. FaceForensics++ or the DeepFake Detection Challenge) and defining a subdomain to work in (e.g. facial/audio deepfakes)
- Implementing a baseline deepfake detection model using pretrained architecture and a detection method.
- Designing and performing controlled evaluation experiments on unseen data and manipulation methods, including cross-dataset testing and domain shift scenarios.

- Investigating the impacts of selected techniques (e.g. augmentation, multimodal, feature extraction) on generalisation and robustness.
- Develop and evaluate a hybrid framework by combining selected detection methods or features to assess whether integration improves performance
- Analyse the model's behavious and identify failure conditions including patterns of misclassification across different settings.
- Compare different approaches and evaluate their robustness under different conditions.
- Conduct ablation studies to isolate the contribution of different model components and features to overall performance and generalisation ability.
- Reaching a conclusion on the limitations of current methods and propose improvements.

State how your project will be aligned with the learning outcomes of your programme of study.

This project aligns with the multiple learning outcomes of the programme across the Academic Content, Disciplinary Skills and Attributes. In terms of Academic Content, the project applies core scientific principles of AI and ML, specifically within Computer Vision and Deep Learning for image and video analysis. It requires the use of statistical methods to evaluate the model performance using quantitative metrics such as accuracy, precision, recall and F1-score. The project also involves the implementation and evaluation of convolutional and transformer-based computational model architectures while assessing their limitations.

For Disciplinary Skills, the project develops my ability to apply and integrate ML techniques in practice, as well as design controlled experiments to evaluate model performance in different conditions. It also involves analysing the model's behaviour in terms of generalising to unseen data and manipulation methods, allowing to identify the limitations of current approaches.

In terms of Attributes, the project develops the ability to engage with AI systems by evaluating their reliability and limitations in real-world applications. This is directly related to deepfake detection where models need to be operating under uncertain and evolving conditions. It also improves research capability since it involves the investigation of an open ended problem and using experimental methods to evaluate and improve model performance. This project also improves my ability to communicate my technical findings effectively through my analysis and presentation of results, including model comparisons, evaluation metrics and failure patterns that have been identified.

METHODOLOGY:

Describe the various steps that you intend to follow for you to achieve your project aims.

The project will follow a structured experimental methodology consisting of the following stages:

1. Literature Review

A review of recent research in deepfake detection will be conducted to identify state-of-the-art methods, common architectures, and known limitations, with a particular focus on generalisation and robustness.

2. Dataset Selection and Preparation

A suitable dataset within a defined subdomain will be selected (e.g. FaceForensics++ or the DeepFake Detection Challenge). The data will be preprocessed, including tasks such as frame extraction, resizing, normalisation, and dataset splitting for training and evaluation.

3. Baseline Model Implementation

Deepfake detection models will be implemented using pretrained architectures (e.g. CNN-based and/or transformer-based models) and initial experiments will establish baseline performance.

4. Experimental Design and Evaluation

Controlled experiments will be designed to evaluate model performance under different conditions, including cross-dataset evaluation, testing on unseen manipulation methods and evaluating performance under varying data conditions using metrics such as accuracy, precision, recall, and F1-score.

5. Investigation of Techniques and Hybrid Framework
A hybrid framework combining selected methods will be developed and evaluated to determine whether integration improves the performance of the model over the baseline.

6. Failure Analysis
Model predictions will be analysed to identify failure conditions and patterns of misclassification. This analysis will include examining whether errors are correlated with specific datasets, manipulation types, or visual/audio artefacts to identify potential sources of model bias or overfitting.

7. Result Analysis and Discussion
The results will be compared across different experimental setups, and conclusions will be drawn regarding the limitations of current methods and the effectiveness of proposed improvements.

PROJECT MILESTONES

Indicate what measurable/tangible components you will produce as part of this project. This may take the form of deliverable document(s) or developmental milestones such as a working piece of software/hardware.

- Completion of literature review and problem definition
- Selection and preprocessing of dataset
- Implementation of baseline deepfake detection model
- Initial experimental results and baseline performance evaluation
- Design and execution of experiments
- Implementation and evaluation of selected techniques and hybrid framework
- Analysis of failure modes and model behaviour
- Completion of full experimental results and comparative analysis
- Demonstrate working prototype or demonstration of model performance
- Draft dissertation report
- Final dissertation submission

REQUIRED KNOWLEDGE/ SKILLS/TOOLS/RESOURCES:

Indicate as far as possible the skills that are required for you to undertake this project. Also include any software, hardware or other tools or resources that you believe you will need.

Technical Knowledge:

- Knowledge of machine learning and deep learning principles, including supervised learning, classification, and model evaluation
- Understanding of computer vision techniques, particularly image and video analysis
- Familiarity with deepfake generation and detection methods, including architectures such as Convolutional Neural Networks (CNNs) and Vision Transformers (ViTs)
- Awareness of generalisation challenges, dataset bias and overfitting in machine learning systems
- Understanding of multimodal learning concepts

Skills:

- Strong proficiency in Python programming
- Experience with Deep Learning frameworks like PyTorch and TensorFlow
- Being able to preprocess and manage image and/or video datasets
- Experience in fine tuning models
- Ability to design controlled experiments to evaluate performance using metrics like accuracy, recall, precision and F1 Score.

- Analytical skills to understand results and identify model limitations

Software Tools:
- PyTorch or TensorFlow for model development
- OpenCV for image and video processing
- Speech processing tools (E.g. Whisper for audio transcription)
- NumPy and Pandas for data handling
- Matplotlib or Seaborn for visualisation
- Jupyter Notebook or Visual Studio Code for development
- Git for version control
- Libraries providing pretrained models (e.g. Hugging Face Transformers)

Hardware Resources:
- Access to GPU resources for model training and experimentation
- Moderate GPU capability suitable for fine tuning pretrained CNN and transformer models.
- Local or Cloud computing platforms
- Adequate storage for datasets and model outputs

Datasets and Other Resources:
- Access to public deepfake datasets, recent academic papers and research publications in deepfake detection and open source implementations of relevant deepfake detection models

Other Skills:
- Being able to critically analyse research papers
- Technical report writing and communication skills
- Time and project management skills
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TIMEPLAN

This can be a GANTT chart submitted with this document or a list of tasks, milestones and deliverables with timings.

Phase 1: Literature Review and Problem Definition (Weeks 1–3)
- Review deepfake detection methods with focus on generalisation and robustness
- Refine research question, hypothesis, and project scope

Phase 2: Dataset Preparation (Weeks 4 - 5)
- Select and preprocess dataset, and define training, validation, and testing setup.
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Phase 3: Baseline Implementation (Weeks 6 - 8)
- Implement and train baseline model and establish initial performance metrics

Phase 4: Experimental Evaluation (Weeks 9 - 12)
- Design and run controlled experiments (e.g. cross-dataset, unseen manipulations)
- Analyse generalisation performance

Phase 5: Model Improvements (Weeks 13 - 16)
- Implement additional techniques and hybrid framework and compare performance against baseline

Phase 6: Failure Analysis (Weeks 17 - 18)
- Analyse misclassifications and identify failure patterns
- Investigate sources of bias and limitations

Phase 7: Finalisation and Writing (Weeks 19 - 20)
- Consolidate results, draw conclusions and finalise dissertation